Fiscal Year:	FY 2026	**Task Last Updated:**	FY 09/30/2025
PI Name:	Clement, Gilles Ph.D.		
Project Title:	Functional Task Tests in Partial Gravity during Parabolic Flight		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	77058-3711	**Congressional District:**	36
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Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per M. Stenger/NASA HHC Element Scientist and C. Ribeiro/NASA HHC (Ed., 1/12/24) NOTE: End date changed to 12/31/2022 per PI; original end date was 9/30/2021 (Ed., 5/3/21)		
Key Personnel Changes/Previous PI:	Per the PI, Sarah Moudy, Ph.D. (KBR, NASA Johnson Space Center) is a Co-Investigator involved in data processing, data analysis, data interpretation, and manuscript writing (Ed., 10/14/25). May 2025 report: Gilles R. Clément, PhD, Principal Investigator, KBR, NASA Johnson Space Center, Houston TX; Timothy Macaulay, PhD, Co-Investigator, KBR, NASA Johnson Space Center, Houston TX; Austin Bollinger, KBR, NASA Johnson Space Center, Houston TX; Scott Wood, PhD, Co-Investigator, NASA Johnson Space Center, Houston TX, Hannah Weiss, KBR, NASA Johnson Space Center, Houston TX.		
COI Name (Institution):	Macaulay, Timothy Ph.D. (KBR/NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Bollinger, Austin (KBR/NASA Johnson Space Center) Weiss, Hannah (KBR/NASA Johnson Space Center) Moudy, Sarah (KBR/NASA Johnson Space Center)		

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Task Description:	Critical mission tasks that are required by crews immediately after landing on a planetary surface include egressing from a seat, jumping, and walking. To define an effective and comprehensive countermeasure strategy for preserving crew performance during exploration-class missions, there is a need to understand how these functional tasks are performed in partial gravity, such as on the Moon or Mars. We will analyze the execution of four critical mission tasks (Seat Egress and Walk, Recovery from Fall and Stand, Jump Down, and Tandem Stance) during the partial gravity and normal gravity phases of parabolic flight by using the same equipment and procedures as those previously used on astronauts returning from International Space Station (ISS) missions and ground-based subjects during axial body unloading. Our hypothesis is that the limits of stability for these activities will be larger when the gravity level is reduced. The largest decreases in performance are expected at the lowest gravity level (0.25 g) because subjects will no longer be able to use the gravitational reference for their perception of upright. Ultimately, this information could be used to assess performance risks and inform the design of countermeasures for NASA exploration-class human missions. The four specific aims include: Specific Aim 1: Seat Egress and Walk. The purpose of this test is to measure the ability to rise from a seated position and walk while avoiding obstacles to test mobility. This test is identical to the Sit-to-Stand and Walk-&-Turn test used for Standard Measures after spaceflight and bed rest. In this test, subjects are requested to rise from a seated position as quickly as possible without using their hands and walk as quickly and safely as possible straight ahead towards a cone (4 m distance), walk around the cone, then return and sit back down in the chair. On the way to and back from the cone, subjects step over a 30-cm obstacle. Two trials will be performed per parabola. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units. Performance metrics include time to complete the trial, turn rate during the turn, obstacle contact, and head-torso coordination. Specific Aim 2: Tandem Stance. The Tandem Stance test is a standard test of static postural stability. This test is similar to the Computerize Dynamic Posturography test performed on astronauts as part of their Medical Requirements and on bed rest subjects as part of the HRP Standard Measures (Postural Equilibrium Control). In this test, at the sound of a tone subjects are instructed to stand upright in a heel-to-toe fashion with their arms crossed on their chest. This test is performed with the eyes open and with the eyes closed. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units. The maximum time (prior to taking a step) as well as the medial-lateral peak-to-peak sway angle (p-p sway) is used to quantify postural stability. Specific Aim 3: Recovery from Fall and Stand. The purpose of this test is to measure the ability to maintain postural control after standing up from a prone position. Impairment in the ability to rise from a prone position is one of the strongest independent risk factors associated with serious fall-related injuries. In this test, subjects rest in a prone position, then stand up as quickly as possible and maintain a quiet standing position. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units. The anterior-posterior and medial-lateral peak-to-peak sway angle (p-p sway) is used to compute the equilibrium score, where 12.5 is the maximum theoretical p-p sway. This test also induces an orthostatic challenge. Therefore, heart rate is collected continuously throughout this test. This cardiovascular data is used to detect potential signs of orthostatic intolerance during this active head-up tilt test. Specific Aim 4: Jump Down. In the Jump Down test, at the sound of a tone subjects perform a two-footed hop from a height of 30 cm onto a force plate that measures the ground reaction forces on landing. After landing, subjects are instructed to remain still on the force plate, in the standing position, with arms at their sides for 10 s. After 10s, subjects will also perform a maximal voluntary lean in one direction to quantify changes in the limits of stability at different g-levels. Two jump-down trials will be performed per parabola. A video camera records each trial and body motion (head and torso) is recorded from triaxial inertial measurement units. Study Participants. Twelve subjects (6 male, 6 female) will be tested during 3 flights of 30 parabolas, including 10 parabolas at 0.25 g, 10 parabolas at 0.5 g, and 10 parabolas at 0.75 g. In addition, each subject will perform all the functional task tests in 1 g during the flight between parabolas when the aircraft flies straight and level. Risk Characterization, Quantification\Evidence. This study will contribute to gap closure by providing information regarding any functional task performance deficits in partial gravity. The dose-response relationship between gravity level and task performance decrement will also help in determining the gravity threshold for these functional tasks. These functional task tests are selected to simulate critical mission tasks that crewmembers may be required to perform when they land on another planet with partial gravity. Countermeasure\Prototype Hardware or Software. This task will contribute to gap closure by determining the gravity threshold for these functional tasks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The functional tasks tests in the proposed study will challenge balance control, a function that is paramount to the efficient completion of critical mission tasks. The vestibular and sensorimotor systems play a fundamental role in balance control, and the functioning of these systems is altered during parabolic flight. Knowledge gained from the proposed study will allow us to characterize the risk of balance impairments in various partial gravity levels, thus ensuring that a more effective and comprehensive countermeasure strategy can be developed for preserving crew performance during exploration-class missions.

Task Progress:	Twelve volunteers were tested during parabolas that produced 0.25 g, 0.5 g, or 0.75 g aboard Novespace's Zero-G aircraft, and at 1 g during level flight intervals between parabolas. Additionally, 14 other subjects were tested using identical procedures in a 1 g laboratory setting. The findings indicate that reduced gravity alters the performance of activities such as settling after standing and navigating around obstacles. As gravity levels decreased, postural instability increased, which increased the time required to stand up, settle, walk, and negotiate obstacles. In addition, there were significantly more falls recorded at lower gravity levels. However, reduced gravity did not affect the time required to settle after jumping down. These results have been published in a peer-reviewed journal article (Clement et al., 2024). Additional analyses of head-trunk coordination showed alterations in magnitude-squared coherence (MSQ) in pitch and velocity root mean square deviation (RMSD) in roll, suggesting increased coordination (or "locking") between the head and trunk at low levels of partial gravity. Our study's findings reveal that partial gravity significantly impacts the execution of maneuvers that require greater reliance on balance control, such as tandem stance, seat egress and walk, and recovery from potential falls. Similar declines in maneuvers demanding high dynamic control of postural equilibrium are also evident in astronauts immediately after their return from space. Therefore, it can be inferred that the decreases in postural control that astronauts will experience upon landing on the Moon or Mars may surpass those observed in our study. Consequently, in-flight countermeasures targeting vestibular and proprioceptive systems will be crucial to sustain crewmembers' balance function, enabling them to successfully execute critical mission tasks. Three additional investigations were performed to further explore the effects of gravity levels and gravity transitions on vestibular function (see section D1 "Changes in approach and reasons for change"). The first was a literature review of potential benefits and human systems integration of parastronauts with bilateral vestibulopathy for spaceflight missions. Upon landing after long-duration spaceflight, astronauts often experience motion sickness and impaired performance of functional tasks. Parastronauts with bilateral vestibulopathy are immune to motion sickness; therefore, these individuals might be better prepared for landing after spaceflight. They have adapted strategies for maintaining balance and orientation without relying on vestibular inputs, potentially making them more stable and less prone to disorientation in microgravity or rotating environments. Their unique adaptations may allow them to perform many mobility tasks more effectively during critical mission phases, such as vehicle egress, when other crew members might be more affected by vestibular issues. While they may not perform all tasks, these parastronauts can excel in specific roles that leverage their unique abilities, contributing to the mission's overall success in specialized capacities. We propose using lunar gravity achieved during parabolic flight and prolonged centrifugation as models to study how functional task performance might be less impaired in parastronauts with bilateral vestibulopathy compared to healthy individuals when landing on the Moon after extended exposure to microgravity (Ramsburg et al., 2025). Data from a separate parabolic flight experiment were analyzed to assess how individuals perceive the amplitude of passive body translation in microgravity and hypogravity. Six blindfolded participants reported their perceived amplitudes of whole-body translations ranging from 25 to 250 cm along the three spatial axes under 0 g, 1 g, and 1.8 g conditions. Results showed that the perceived amplitudes of these translations were accurate in 1 g. However, subjects significantly underestimated distances in 0 g and overestimated them in 1.8 g. These results have been published in a peer-reviewed journal article (Clement et al., 2025a). These findings suggest that alterations in gravitational cues disrupt the vestibular system's ability to provide accurate information on body movement, leading to altered motion perception. More specifically, in 1 g, the otolith organs detect both gravitational and inertial accelerations, allowing the brain to distinguish between gravity and self-motion. In 0 g, however, the otoliths respond solely to inertial acceleration, leading to errors in motion perception. Conversely, in hypergravity conditions–such as the 1.8 g pull-up phase of a parabolic flight–translation amplitude is overestimated. This suggests that the brain relies on an expected ratio between inertial and gravitational forces. When gravity increases, acceleration is perceived as stronger, resulting in an overestimation of motion. Since other factors, such as proprioceptive inputs and temporal cues are also impacted by altered gravity levels, external tools like visual and proprioceptive aids must be developed to help astronauts maintain spatial orientation during critical spaceflight operations. A final separate analysis was performed to investigate how short-duration spaceflight affects private astronauts' performance of mission-critical functional tasks that challenge balance and locomotor control systems shortly after they return to Earth. Ten astronauts were assessed while they performed three functional tests (sit-to-stand, tandem walk, and walk-and-turn) before spaceflight and a few hours after returning from missions lasting from 4 to 21 days (Mean ± SD; 10.4 ± 7.4 days). Their performance was compared to that of 36 astronauts who returned from long-duration missions lasting from 5 to 12 months (6.8 ± 1.7 months). For all tasks, there were no significant differences in preflight performance between short and long duration groups. Shortly after returning from spaceflight, the short-duration astronauts had difficulty standing, walking, and turning around obstacles, and they experienced terrestrial readaptation motion sickness. However, the performance of these functional tasks was less impacted after short-duration missions than long-duration missions. These results have been published in a peer-reviewed journal article (Clement et al., 2025b). The decrements in postural and locomotor control after short-duration ISS missions are comparable to those seen after Space Shuttle missions of similar length, suggesting that these effects are not dependent on the spacecraft type or the launch and re-entry characteristics. Although individual responses vary, the overall performance impairments are generally less severe after short-duration missions compared to long-duration missions. Longer missions involve more substantial physiological changes—such as greater losses in muscular strength and proprioception function — that can compound sensorimotor disruptions and further impact performance. As a result, shorter missions may require fewer or less-intensive countermeasures than long-duration flights, allowing for faster recovery, simplified rehabilitation, and more efficient use of resources during re-adaptation to Earth's gravity.
Bibliography Type:	Description: (Last Updated: 10/10/2025)
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